Aayush Mittal

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Summary

Full-stack AI engineer (**Python, TypeScript, React, MCP**). Built **LACE**, an open-source, local-first agent memory engine (**RAG**, knowledge graph) that AI agents use over MCP. Delivered **5+ real-world projects** for **real clients** with **positive client reviews**. **4+ hackathon** qualifier. **B.Tech ECE**, IIIT Kota.

Education

Indian Institute of Information Technology, Kota 2024 – 2028
B.Tech, Electronics and Communication Engineering ***CGPA: 9.00/10.00***

Projects

LACE – Local AI Context Engine | *Python, MCP, ChromaDB, NetworkX, SQLite, LLM APIs* Feb 2026 – Present

- Built a local-first memory engine for AI coding agents, exposing **9 tools and 5 resources over MCP** to clients like Claude Desktop and Cursor; memories are stored as Markdown in an Obsidian-compatible vault.
- Designed a **RAG retrieval pipeline** ranking memories on 6 signals (vector similarity, tags, graph, co-retrieval, recency, confidence) over ChromaDB embeddings and a NetworkX knowledge graph.
- Decoupled writes with a SQLite (WAL) queue: $<$**5 ms enqueue**, async LLM-as-judge extraction, two-tier cosine dedup (drop $>$95%, merge 85–95%).
- Tested with **pytest**, documented design decisions via **ADRs**, MIT-licensed: github.com/AayushM0/LACE

Global Agrawal Directory | *Next.js 15, React 19, TypeScript, PostgreSQL, Supabase, Pusher* 2026 – Present

- Built on **Next.js 15 (App Router), TypeScript, PostgreSQL and Supabase**; in production at **maharajaagrasenfoundation.com** for **Maharaja Agrasen Foundation, Singapore** (real client) with **400+ registered accounts** across **7+ countries** and **4 live modules** (directory, business network, matrimonial portal, jobs & careers).
- Built a **5-step household registration wizard** on **Supabase OTP auth** with **bcryptjs**-hashed verification and field-level privacy toggles (DOB, contact, photo) stored in PostgreSQL.
- Implemented member-level **OTP self-verification**: a verified member's row gets an `owner_locked` flag in PostgreSQL, transferring edit rights from the household head to that member.
- Built **PostgreSQL-backed faceted search** across **18 Gotras**, city and geo radius, with an admin moderation queue (approve/reject, soft-delete) gating new households before they go live.
- Integrated **Pusher** for real-time UI updates, **react-pdf/renderer** for PDF generation, and **Docker Compose** for local PostgreSQL.

Experience

SMG Electric Scooters Ltd. Dec 2024 – Jan 2025
Winter Intern | *React, Vite, Node.js, TypeScript, MongoDB, Redis, Socket.IO, Docker* *India*

- Built a full-stack ride-hailing platform (Uber/Ola-style) for an internal EV-mobility project: booking flow, rider–driver matching, and live ride status across **5+ pages** (React + Vite, deployed on Vercel).
- Built the **Node.js/Express/TypeScript** API (**20 REST endpoints**) on **MongoDB** with **JWT + phone-OTP auth**, rider/driver/admin roles, and Google Maps API for routes and ETA.
- Real-time matching over **Socket.IO** (separate rider and driver namespaces); **Redis** for driver location/status caching, OTP expiry and ride-acceptance locks; Dockerized with Render deploy config.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, C++, Java
AI / Agents: MCP, RAG, embeddings, vector search (ChromaDB), knowledge graphs, LLM APIs, LLM-as-judge
Frontend: React, Vite, Tailwind CSS, PWA (Service Workers, IndexedDB), HTML/CSS
Backend & Data: Node.js, Express, Fastify, FastAPI, Next.js, REST APIs, WebSockets (Socket.IO, Pusher), JWT, RBAC, Supabase, PostgreSQL, MongoDB, Redis, SQLite
Tools: Git, GitHub Actions, Docker, Linux, pytest

Leadership

Web Development & Web3 Lead, Codebase – IIIT Kota Aug 2025 – Aug 2026
Campus Mantri (Official Campus Representative), GeeksforGeeks Jun 2026 – Dec 2026

Achievements

Prize Winner, ZINNOVATIO 3.0 (2025) · SMG Award · 4+ hackathon qualifier (SHIELD 1.0, Odoo $\times$ GCET: built RentalFlow)